MILITARY SPECIFICATION

SWITCHING UNIT, AUDIO SA-1489/ARC

AND MOUNT MT-3749/ARC

This specification has been approved by the

Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be used in conjunction with the Romeo-8 Equipment to provide signal switching, audio amplification and red-black isolation for the purpose of making the Romeo-8 electrically interchangeable with the Juliet-28 in certain installations.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

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MIL-T-18303 Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

MIL-N-18307 Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

MIL-A-23595 Amplifiers, Audio Frequency AM-3497A/A and AM-4326/A

Naval Air Systems Command

AR-5 Microelectronic Devices Used in Avionic Equipment, Procedures for

AR-8 Versatile Avionic Shop Test System/ Avionics Systems Compatibility, General Requirements for

AR-9 Versitile Avionics Shop Test System Test Program

AR-10 Maintainability of Avionics Equipment and Systems, General Requirements for

STANDARDS

Military

MIL-STD-704 Electric Power, Aircraft Characteristics and Utilization of

MIL-STD-781 Reliability Tests, Exponential Distribution

MIL-STD-756 Reliability Prediction

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

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3.2 Parts and Materials - In the selection of parts and materials, fullfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modulator Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of AR-5.

3.2.3 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules (see Specification AR-10). Maintenance modules shall normally be considered repairable.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

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3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 4 pounds.

3.3.2 Reliability of Switching Unit, Audio SA-1489/ARC -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 5000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 20,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 5000 hours of mean (operating) time between failures as demonstrated by using the analytical methods specified by MIL-STD-756 and the Tests of paragraph 4.4.3.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that protion of the connectors attached to the cables shall not be supplied as part of the equipment.

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3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181. In addition, Red/Black isolation shall be at least 80 db.

3.3.6 Provisions for Maintainability - Built-in test equipment, construction and packaging, and provisions for test points shall be as specified in Specification AR-10. Equipments are required to be compatible with the Versatile Automatic Shop Test System (VAST) in accordance with Specification AR-8. When required, VAST Test Programs shall be furnished in accordance with Specification AR-9.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ±0.5 vdc

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment except as modified herein:

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3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.3.10 Warm-Up Time - The time required for the equipment to warm up prior to operation shall not exceed 10 seconds.

3.3.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704. Limits shall be as specified therein except emergency, steady state d-c voltage limits shall be 21 to 29 vdc. The power required shall not exceed the specified amount.

(1) DC Power 28 v, Category "B", 0.4 amp

3.3.11.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment. Degraded performance will also be permitted for voltage transients during abnormal electric system operation simulated by 80 vdc applied for 5 seconds in lieu of transient voltage specified in paragraph 5.2.3 of MIL-STD-704. Operation shall return to normal within 5 seconds with no resulting damage to the equipment due to this abnormal electric-system operation.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The SA-1489/ARC shall be used in combination with the Romeo 8 Equipment. The switching unit allows the Romeo 8 to be

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electrically interchangeable with the Juliet 28 in selected installations. The interface connectors of the switching unit are the same type as used in the Juliet 28.

The Switching Unit provided audio amplification in "C" mode, audio switching from "P" mode to "C" mode and functions as a junction box for the various Romeo 8 input and output requirements with red-black isolation as required.

In "C" mode, with push-to-talk key closed, the switching unit privides microphone current to a carbon microphone or voltage for an in-line amplifier. The microphone information is attenuated and fed to the Romeo 8 by the switching unit. This information is then returned to the switching unit where it passes through a low pass filter and is applied to a radio modulator input. The information from the Romeo 8 is also returned to the switching unit audio amplifier and routed through relay contacts to the headphones as a transmitter sidetone signal.

With the push-to-talk key open, the system is in the receive mode. A "C" signal from a receiver is routed through the switching unit to the Romeo 8. The audio output from the Romeo 8 is then amplified by the switching unit and applied to the headphones.

In "P" mode, the relays in the switching unit switch the microphone signal directly to the transmitter modulator and connect the headphones directly to the radio audio output.

In "C" mode, the switching unit accepts and amplifies a signal from the radio guard receiver. This auxiliary guard signal is heard by the operator in the headphone.

The Romeo 6 and switching unit swithcing functions are operated

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by the Juliet 28 control unit. The zeroize function is routed through the switching unit directly to the Romeo 8.

3.4.2 Audio Characteristics in "C" Mode

3.4.2.1 Headset Audio Output Characteristics:

3.4.2.1.1 Headset Audio Output Capability - A headset audio output shall be provided which has two inputs isolated from each other:

(1) headset audio input (HAUD)

(2) guard audio input

When a signal having a level of 2.5 vrms at 1000 Hz is applied to the headset audio input, the unit shall be capable of delivering a headset audio output power of at least 250 milliwatts.

When a signal having any level between 0.15 and 0.4 vrms at 1000 Hz is applied to the guard audio input, the unit shall be capable of delivering a headset audio power output of at least 250 milliwatts.

Under any combination of environmental conditions, the headset audio output level shall not degrade more than 3 db at 1000 Hz.

3.4.2.1.2 Headset Audio Output Fidelity - The headset audio output response shall not vary more than ±3 db from 300 to 3000 Hz relative to the response at 1000 Hz.

The total harmonic content of the headset audio output shall not exceed 10% at the 250 milliwatt level at 1000 Hz.

The headset audio output shall be attenuated outside the frequency range of 300 to 3000 Hz to the maximum practicable extent when guard audio input is applied, without recoursd to inclusion of large and/or heavy filter components in the unit.

3.4.2.1.3 Headset Audio Output Adjustment Range - The unit shall provide

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a headset audio input level adjustment capable of maintaining the headset audio power output at a 250 milliwatt level when the headset audio input is increased by any level up to +20 db.

The unit shall provide a guard audio input level adjustment capable of maintaining the headset audio power output at a 250 milliwatt level when the guard audio input is increased by any level up to 20 db.

3.4.2.2 Transmit Wideband Data Output Characteristics - From a single wideband data input, the unit shall provide two wideband data outputs:

(1) Low level wideband data output

(2) High level wideband data output

DC blocking shall be provided between the input and both outputs.

3.4.2.2.1 Transmit Wideband Data Output Capability - When a signal level of 4.25 vrms (open circuit) at 1000 Hz from a 100 ohm source is applied to the wideband data input, the transmit low level wideband data output shall be 0.4 ±0.07 vrms and the transmit high level wideband data output shall be 3.75 ±.2 vrms when both outputs are terminated in their proper loads at the same time.

3.4.2.2.2 Transmit Wideband Data Output Fidelity - The frequency response of both the transmit high level wideband data output and the transmit low level wideband data output shall not vary more than 0 to -4 db from 300 Hz to 25 KHz relative to the response at 1000 Hz. At frequencies above 25 KHz, the outputs shall be attenuated approximately 6 db per octave.

3.4.2.3 Mike Audio Output Characteristics - The unit shall be capable of delivering a mike audio output of 100 millivolts +20 mv, -30 mv when an audio signal of 0.25 vrms, 1000 Hz is applied to the mike audio input.

3.4.3 +18 vdc Output - The unit shall provide a d-c output of within -1 to +2V of +18V into an 18K ohms resistive load.

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3.4.4 Audio Switching - Three audio lines shall be switched in the unit when +28 vdc supply voltage is applied between the P/C control line and ground, placing the unit in "P" mode.

After being switched to "P" mode, the circuits shall provide a d-c path from the inputs to the outputs with less than 1/2 db loss from 300 Hz to 20,000 Hz.

3.4.4.1 Mike Audio In - In "P" mode, the mike audio in shall be switched to the mike audio to modulator output. In the "C" mode only, a d-c voltage shall be applied to the mike audio input to provide 30 to 50 ma current for a carbon microphone or voltage for an in-line amplifier in accordance with Figure 2 of MIL-A-23595.

3.4.4.2 Headset Audio Out - In "P" mode, the headset audio output shall be switched to the Headset audio from the UHF Comm input.

3.4.4.3 Receiver Wideband Data In - In "P" mode, the receiver wideband data in shall be switched from the Romeo 8 to the auxiliary audio to the ADF output.

3.4.5 Through Lines - The unit shall provide a d-c path from input to output with less than 1/2 db loss at 1 KHz for the following functions.

(1) Mike shield
(2) Mike ground
(3) +27.5 vdc
(4) P/C control
(5) Zeroize
(6) Push-to-talk

Grounds, shield grounds and d-c return lines shall be provided as required by the procuring agency.

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3.5 Detail Requirements

3.5.1 Audio Switching Unit SA-1489/ARC - The audio switching unit shall meet the following requirements:

3.5.1.1 Function - This unit shall provide signal switching, audio amplification and red/black isolation necessary to provide a means of making the Romeo 8 electrically interchangeable with the Juliet 28 in UHF/AM installations.

3.5.1.2 Form Factor - The overall maximum dimensions of the unit shall not exceed 2-7/8 inches in height, 7-3/16 inches in width and 5-7/16 inches in depth including the connectors. The unit shall be capable of being mounted in an aircraft by means of Mount MT-3749/ARC.

3.5.1.3 Weight - The weight of the switching unit shall not exceed 3 lbs. 9 ozs.

3.5.1.4 Controls - The unit shall have the following screw-driver adjustments located below connectors J3 and J4 respectively:

Designation	Function
HDST AUD ADJ R2	Adjusts the audio level from the Romeo 8 Equipment to the switching unit audio amplifiers.
GUARD ADJ R19	Adjusts the guard audio level from the UHF transceiver to the switching unit audio amplifiers.

3.5.1.5 Electrical Connections - Connections to external circuits shall be provided as follows:

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Receptacle Reference Designation	Receptacle Type	Function
J1	MS-3122-E-12-10P	Inlet for Red Signals
J2	MS-3122-E-16-26PW	Outlet for Black signals
J3	MS-3122-E-12-10PX	Outlet for Red signals
J4	MS-3122-E-16-26SW	Inlet for Black signals

3.5.1.6 Electrical Performance Characteristics - See paragraph 3.4.

3.5.1.7 Finish - The switching unit shall have a gray texture finish.

3.5.1.8 Terminations and Input Impedances - In cipher mode, the inputs and outputs of the unit shall be terminated or have input impedances as listed below.

3.5.1.8.1 Mike Audio In - The mike audio in input impedance shall be 115 ohms ±20% mainly resistive at audio frequencies.

3.5.1.8.2 Headset Audio Output - The headset audio output may be terminated with any one of three resistive terminations, 600 ohms, 300 ohms or 150 ohms by selection of the proper tap on the impedance matching audio output transformer.

3.5.1.8.3 Transmit High Level Wideband Data Output - The transmit high level wideband data output shall be terminated with a 2000 ohm resistive load.

3.5.1.8.4 Transmit Low Level Wideband Data Output - The transmit low level wideband data output shall be terminated with an 82 ohm resistive load.

3.5.1.8.5 Guard Audio In - The guard audio in input impedance shall be 600 ohms ±20% at audio frequencies.

3.5.1.8.6 Headset Audio In - The headset audio in input impedance shall be 600 ohms ±20% at audio frequencies.

3.5.1.8.7 Mike Audio Output - The mike audio output shall be terminated with a 50,000 ohm resistive load.

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3.5.2 Mount MT-3749/ARC - The mount shall meet the following requirements.

3.5.2.1 Function - The mount shall provide all mechanical support required by the switching unit SA-1489/ARC.

3.5.2.2 Form Factor - The overall dimensions of the mount shall not exceed 1-1/4 inches in height, 7-5/16 inches in width and 6-5/8 inches in depth.

3.5.2.3 Weight - The weight of the mount shall not exceed 7 ounces.

3.5.3 Romeo-8 Adapter Unit - When the audio switching unit SA-1489/ARC is mounted in the Mount MT-3749/ARC, the Romeo-8 Adapter Unit shall meet the following requirements.

3.5.3.1 Form Factor - The size of the unit when the audio switching unit SA-1489/ARC is mounted in the mount MT-3749/ARC shall not exceed 3-5/16 inches in height, 7-5/16 inches in width, and 6-5/8 inches in depth including the connectors. A minimum height of 4-1/8 inches height shall be required to remove the SA-1489/ARC from the MT-3749/ARC.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all

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applicable requirements:

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts, this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

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4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

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4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

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(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Production

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Production - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

*4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

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The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction shall be used. During the 6 hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

*NOTE: This test shall not be required if a reliability test is selected that requires a burn in test for all equipment.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction therof

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*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - The equipment shall be tested in accordance with the Qualification Phase of the reliability assurance test requirements of MIL-STD-781.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conduted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

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4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
D.C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed

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by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-R-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are

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available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Test Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The audio switching unit is intended for use with a Romeo 8 Equipment in certain installations to provide an equipment electrically interchangeable with the Juliet 28.

6.2 Test Values - Normal and limiting values of performance data

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shall be determined at input voltages of 27.5 ± 0.5 VDC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be heald to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater that that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicabe subsidiary specifications are in conflict, the following precedence shall apply:

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(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced herein. All referenced specifications shall apply to the extent specified.

6.6 The (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

Remote control Unit C-8057/ARC

Romeo-8 Equipment

UHF Transceiver Modified for Juliet-28 compatibility

Various Aircraft Intercom

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